MODULE-IV
MANUFACTURING PROCESS

Types of manufacturing processes

Manufacturing – Overview

- Manufacturing is the process of converting raw materials into finished products through physical, chemical, or mechanical processes.
- It adds value to raw materials by shaping, forming, assembling, and finishing them into useful goods.
- Objectives of Manufacturing
- To produce goods of desired shape, size, and quality
- To improve product performance and durability
- To achieve mass production with minimum cost and time
- To meet industrial and consumer needs

Types of Manufacturing Processes

1) Casting Processes

- In casting, molten metal is poured into a mould and allowed to solidify.
- Examples
- Sand casting
- Die casting

2) Forming Processes (Metal Forming)

- Material is shaped by plastic deformation without adding or removing material.
- Examples
- Forging
- Rolling

3) Machining Processes (Material Removal)

- Material is removed in the form of chips to achieve the desired shape and dimension.
- Examples
- Turning
- Milling
- Drilling

4) Joining Processes

- Two or more parts are permanently or temporarily joined.
- Examples
- Welding (arc, MIG, TIG)
- Brazing and soldering

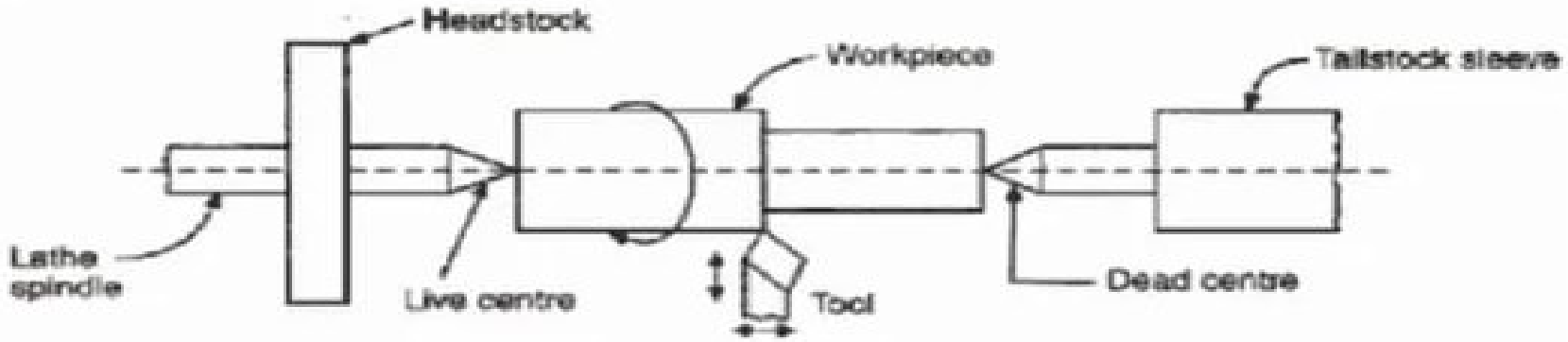
5) Additive Manufacturing (3D Printing)

- Material is added layer by layer to build a product from CAD data.
- Examples
- Fused Deposition Modeling (FDM)
- Selective Laser Sintering (SLS)

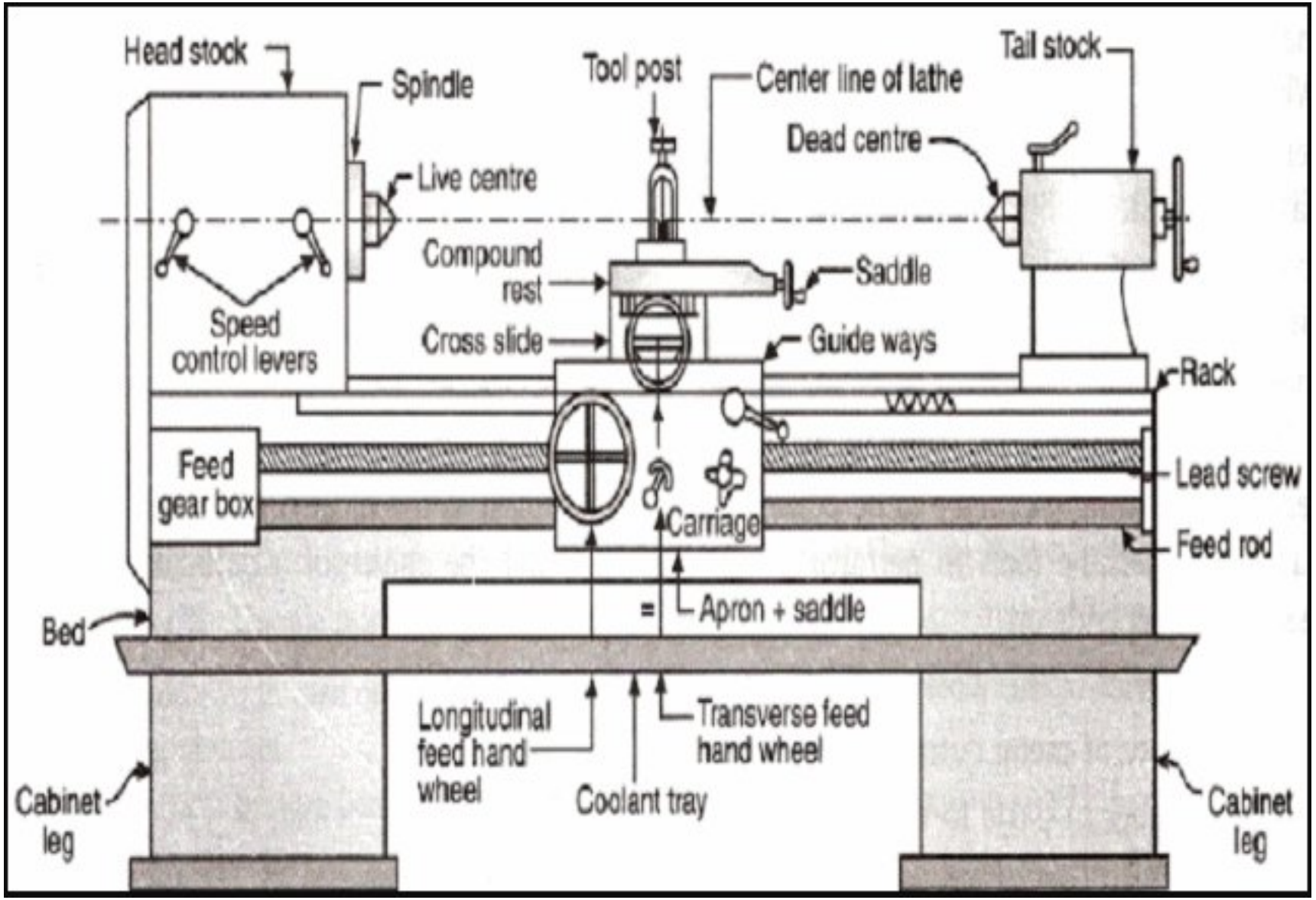
Principle of working of a center lathe

PRINCIPLE OF LATHE MACHINE

THE LATHE IS A MACHINE TOOL WHICH HOLDS THE W/P B/W TWO RIGID & STRONG SUPPORTS CALLED CENTRES OR IN A CHUCK OR IN FACE PLATE WHICH REVOLVES. THE CUTTING TOOL IS HELD AND FED AGAINST THE REVOLVING WORK.CUTTING TOOL FED EITHER PARALLEL OR AT RIGHT ANGLES TO THE AXIS OF W/P. OR MAY ALSO AT AN ANGLE



Different parts of Lathe



Different parts of Lathe

1) Bed: The bed is a heavy, rugged casting in which are mounted the working parts of the lathe. It carries the headstock and tail stock for supporting the work piece and provides a base for the movement of carriage assembly which carries the tool.
2) Legs: The legs carry the entire load of machine and are firmly secured to floor by foundation bolts.
3) Headstock: The headstock is clamped on the left hand side of the bed and it serves as housing for the driving pulleys, back gears, headstock spindle, live centre and the feed reverse gear. The headstock spindle is a hollow cylindrical shaft that provides a drive from the motor to work holding devices.
4) Gear Box: The quick-change gear-box is placed below the headstock and contains a number of different sized gears.
5) Carriage: The carriage is located between the headstock and tailstock and serves the purpose of supporting, guiding and feeding the tool against the job during operation. The main parts of carriage are:

    a). The saddle is an H-shaped casting mounted on the top of lathe ways. It provides support to cross-slide, compound rest and tool post.

    b). The cross slide is mounted on the top of saddle, and it provides a mounted or automatic cross movement for the cutting tool.

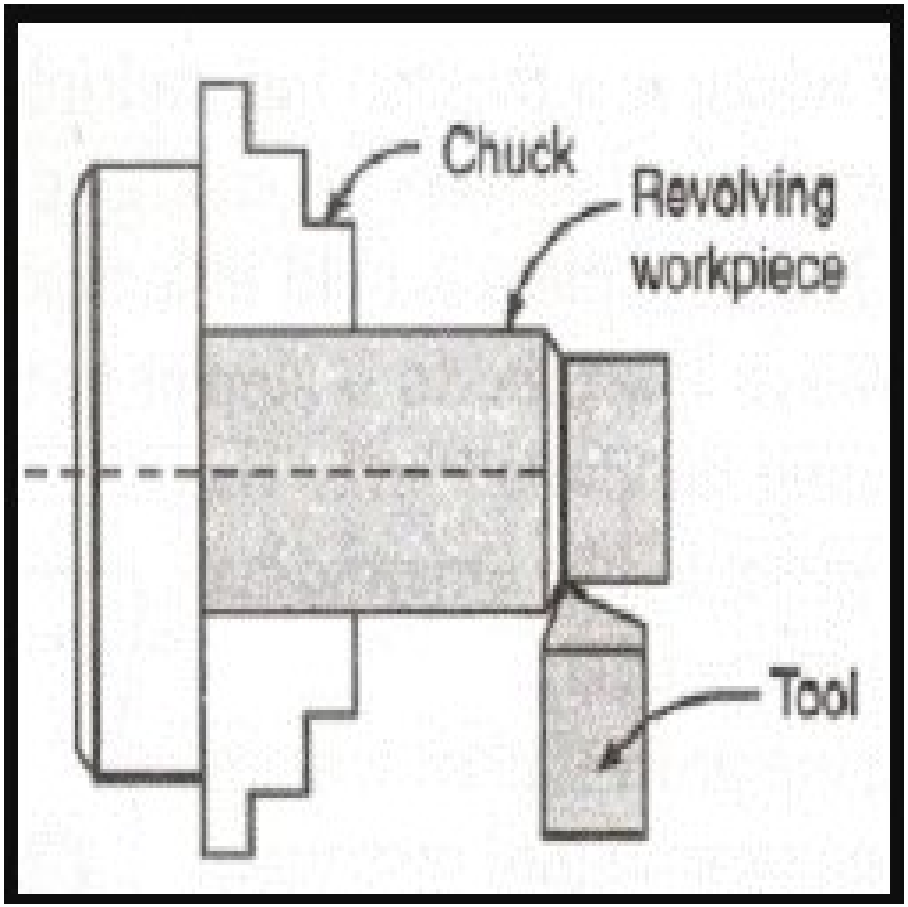
    c). The compound rest is fitted on the top of cross slide and is used to support the tool post and the cutting tool.

    d). The tool post is mounted on the compound rest, and it rigidly clamps the cutting tool or tool holder at the proper height relative to the work centre line.

    e). The apron is fastened to the saddle and it houses the gears, clutches and levers required to move the carriage or cross slide. The engagement of split nut lever and the automatic feed lever at the same time is prevented she carriage along the lathe bed.

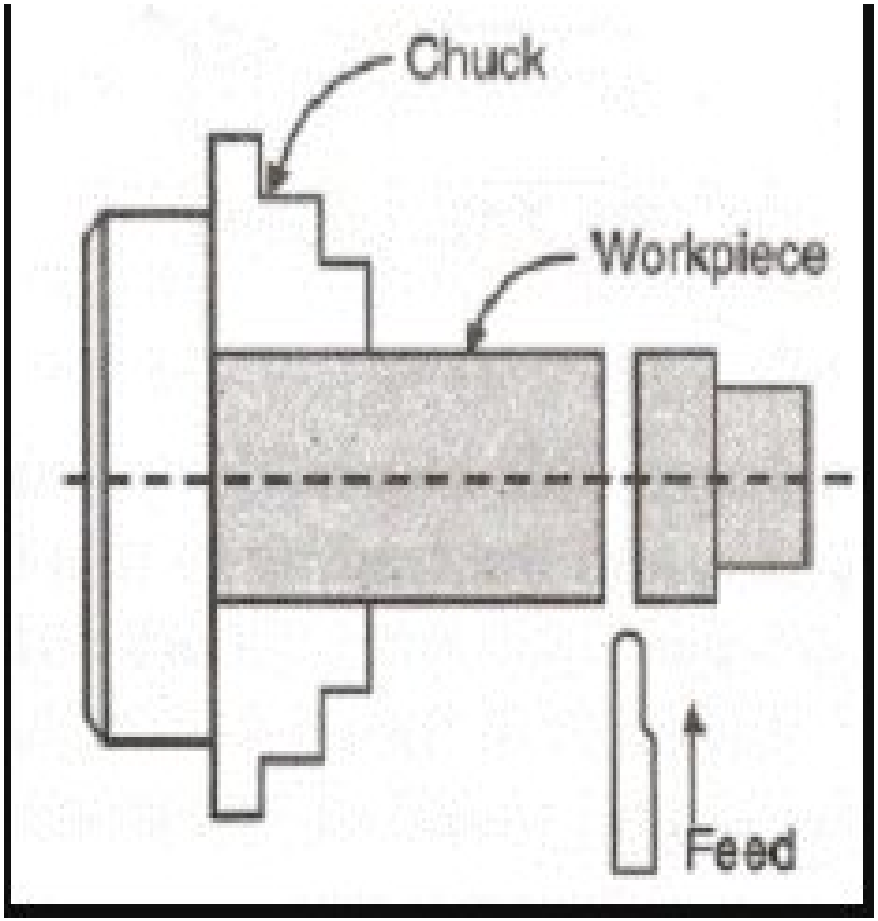
6) Tailstock: The tailstock is a movable casting located opposite the headstock on the ways of the bed. The tailstock can slide along the bed to accommodate different lengths of work piece between the centers. A tailstock clamp is provided to lock the tailstock at any desired position. The tailstock spindle has an internal taper to hold the dead centre and the tapered shank tools such as reamers and drills.

Plain Turning



- Plain turning is the operation of removing excess amount of material from the surface of a cylindrical job.
- As the tool moves, it removes material from the outer diameter of the workpiece, reducing its size to a uniform diameter — this is known as plain turning.

Facing



- The facing is a machining operation by which the end surface of the work piece is made flat by removing metal from it.
- The tool moves from the outer edge toward the center of the face of the workpiece, producing a smooth, flat surface that is perpendicular to the axis of rotation.

Knurling

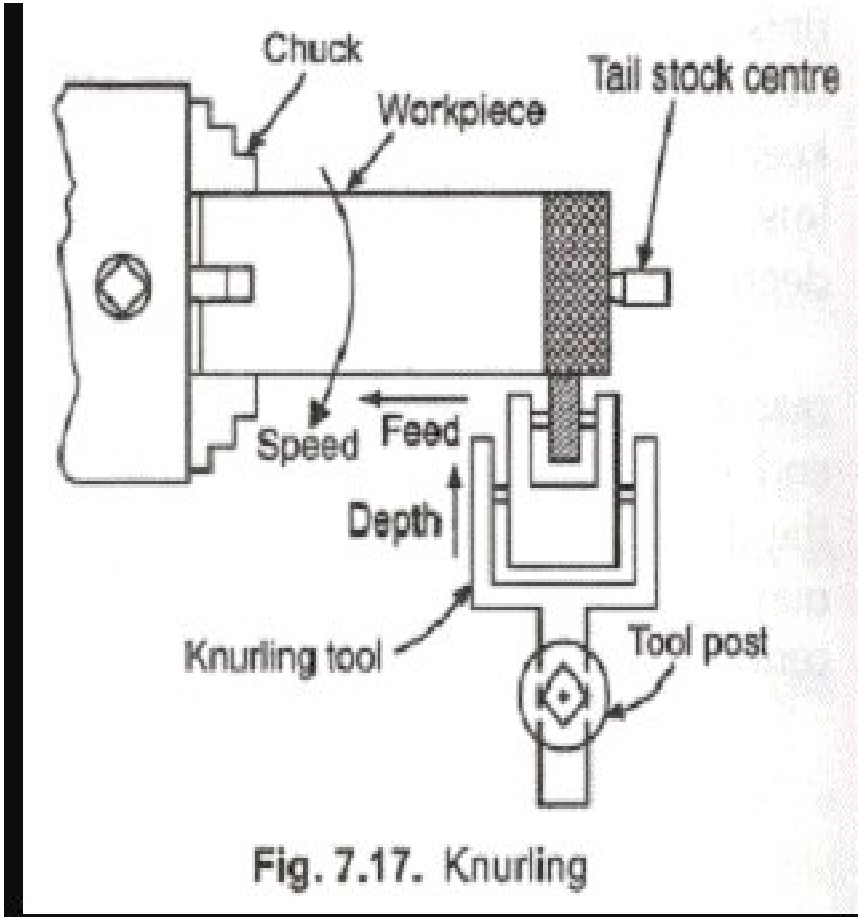


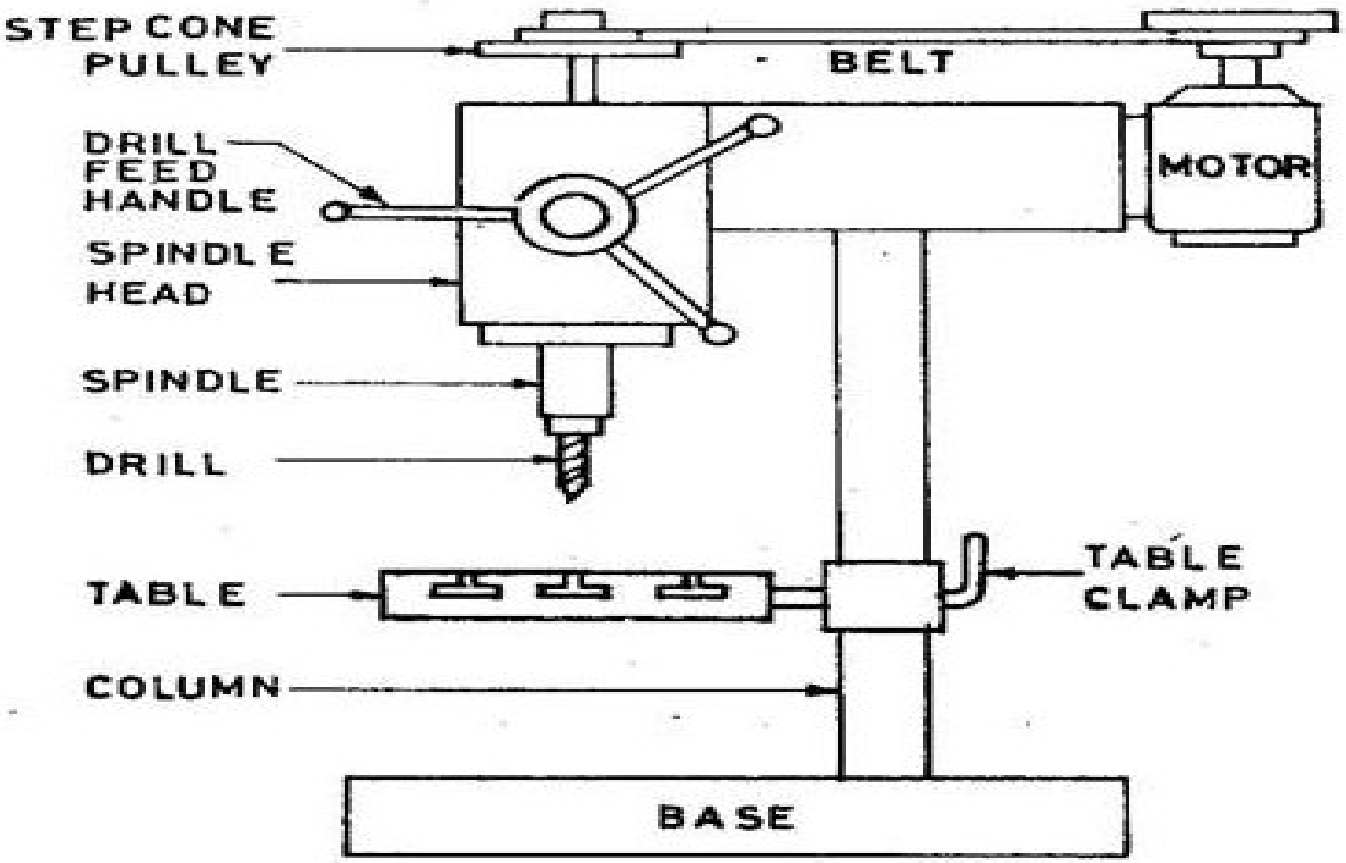
Fig. 7.17. Knurling

- The knurling is a process of embossing (impressing) a diamond-shaped or straight-line pattern into the surface of workpiece.
- Tool: A knurling tool with hardened patterned rollers presses against the rotating workpiece.
- As the rollers are fed against the surface, they displace material to form the raised patterns.
- Knurling is essentially a roughening of the surface and is done to provide a better gripping surface.

Working of simple drilling machine

- The drilling machine or drill press is one of the most common and useful machine employed in industry for producing forming and finishing holes in a workpiece.
- Working principle: The rotating edge of the drill exerts a large force on the workpiece and the hole is generated. The removal of metal in a drilling operation is by shearing and extrusion.

Construction of Drilling Machine



Construction of Drilling Machine

- The main operating parts of a drilling machine are Base, Column, Table, and Drill Head.

1) Base: The base is a heavy casting that supports the machine structure; it provides rigid mounting for the column and stability for the machine. The base is usually provided with holes and slots which help to Bolt the base to a table or bench and allow the work-holding device or the workpiece to be fastened to the base.
2) Column: The column is a vertical post that Column holds the worktable and the head containing the driving mechanism. The column may be of round or box section.
3) Table: The table, either rectangular or round. Drill machine/press in shape supports the workpiece and is carried by the vertical column. The surface of the table is 90-degree to the column and it can be raised, lowered and swiveled around it. The table can be clamp/hold the required the workpiece. Slots are provided in most tables to allow the jigs, fixtures or large work pieces to be securely fixed directly to the table.
4) Drilling Head: The drilling head, mounted close to the top of the column, houses the driving arrangement and variable speed pulleys. These units transmit rotary motion at different speeds to the drill spindle. The hand feed lever is used to control the vertical movement of the spindle sleeve and the cutting tool.

Drilling

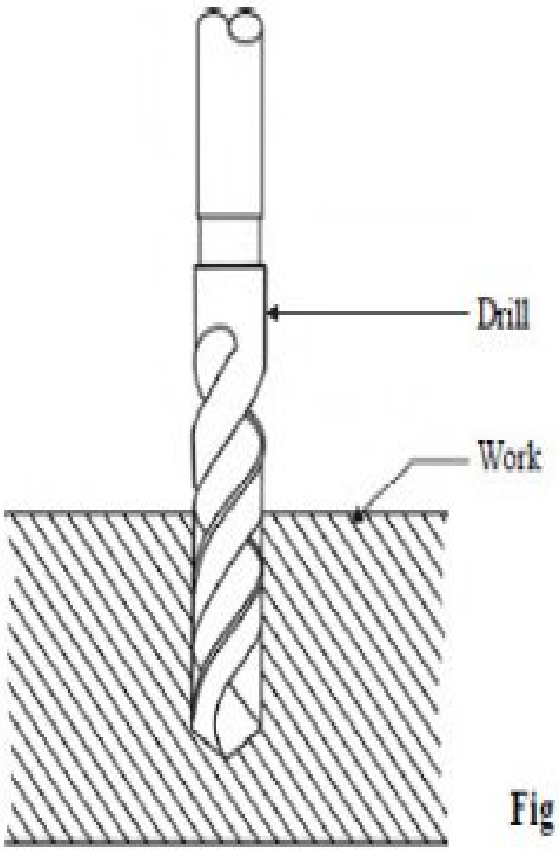


Fig 2.17 Drilling

- Drilling is the operation of producing a cylindrical hole in the workpiece.
- Drilling is the operation of producing a hole by removing metal from a solid mass by the rotating edge of a cutting tool known as drill.

Reaming

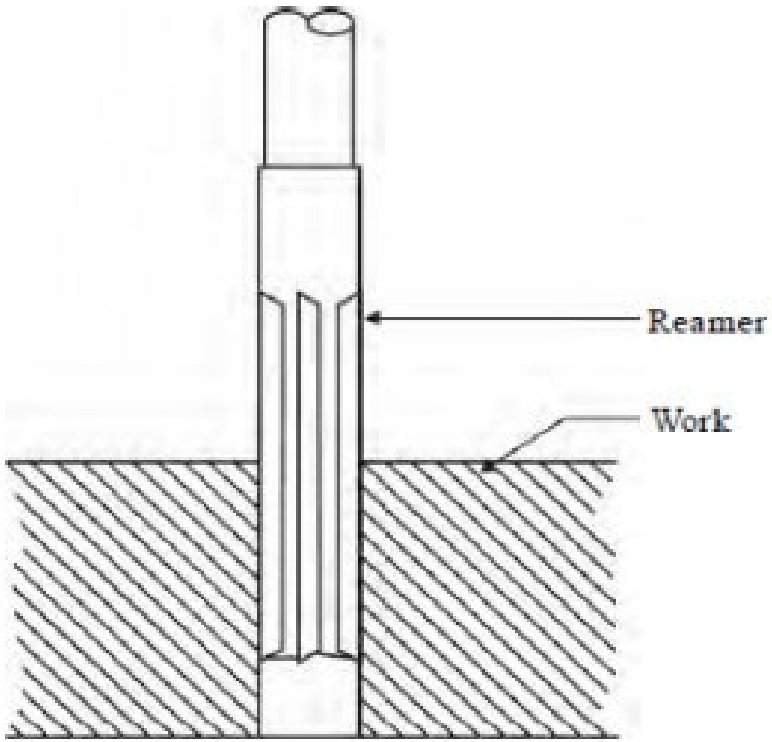


Fig 2.18 Reaming

- The holes that are produced by drilling are rarely straight and cylindrical in form. The reaming operation finishes and sizes the hole already drilled into the workpiece.
- To produce accurate and smooth holes, the drilled holes are reamed by a tool called reamer.

Tapping

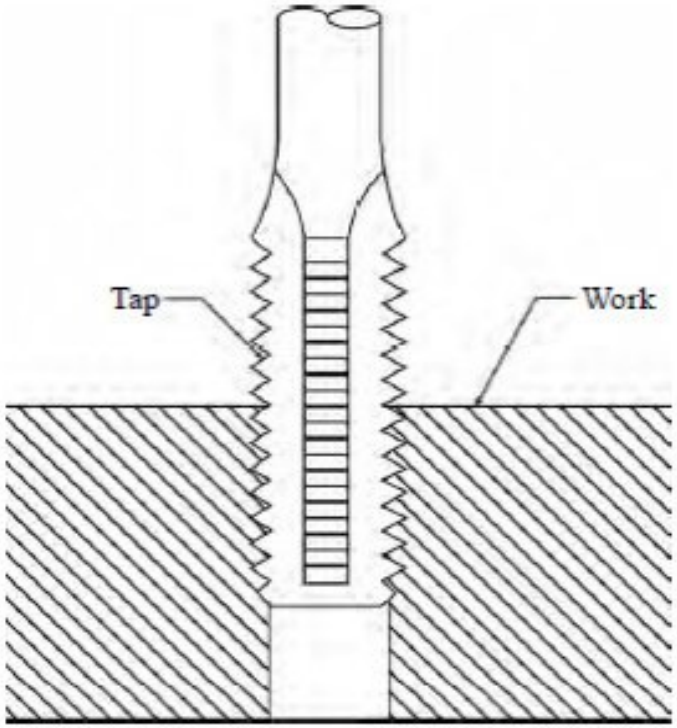
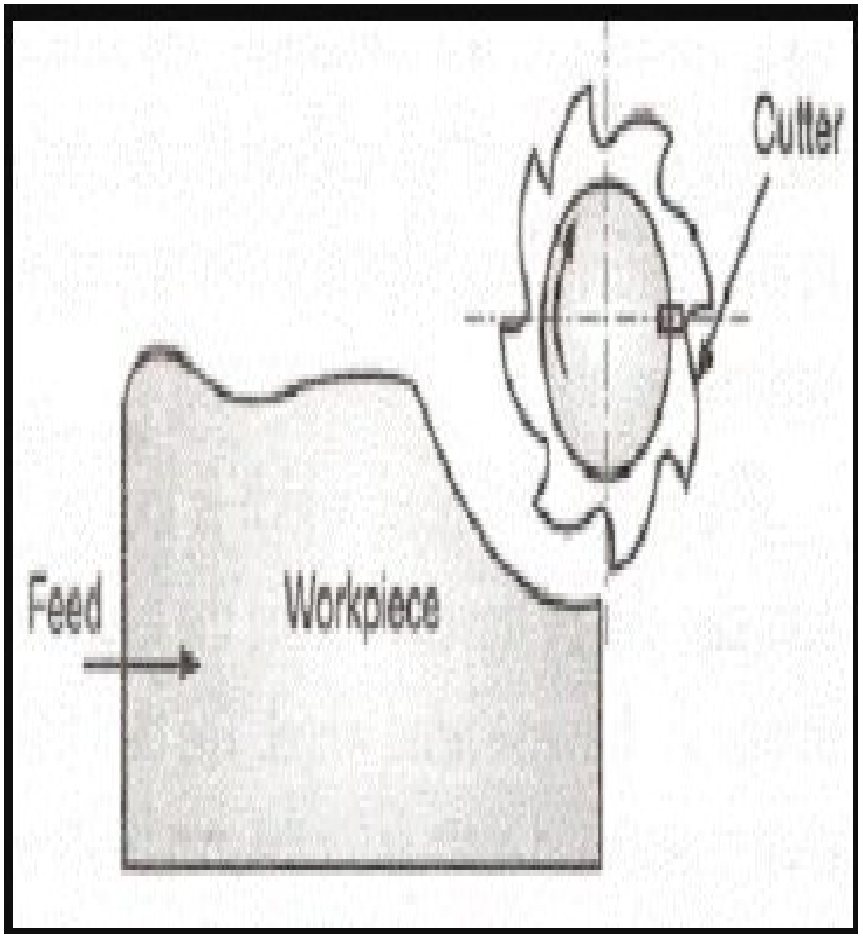


Fig 2.23 Tapping

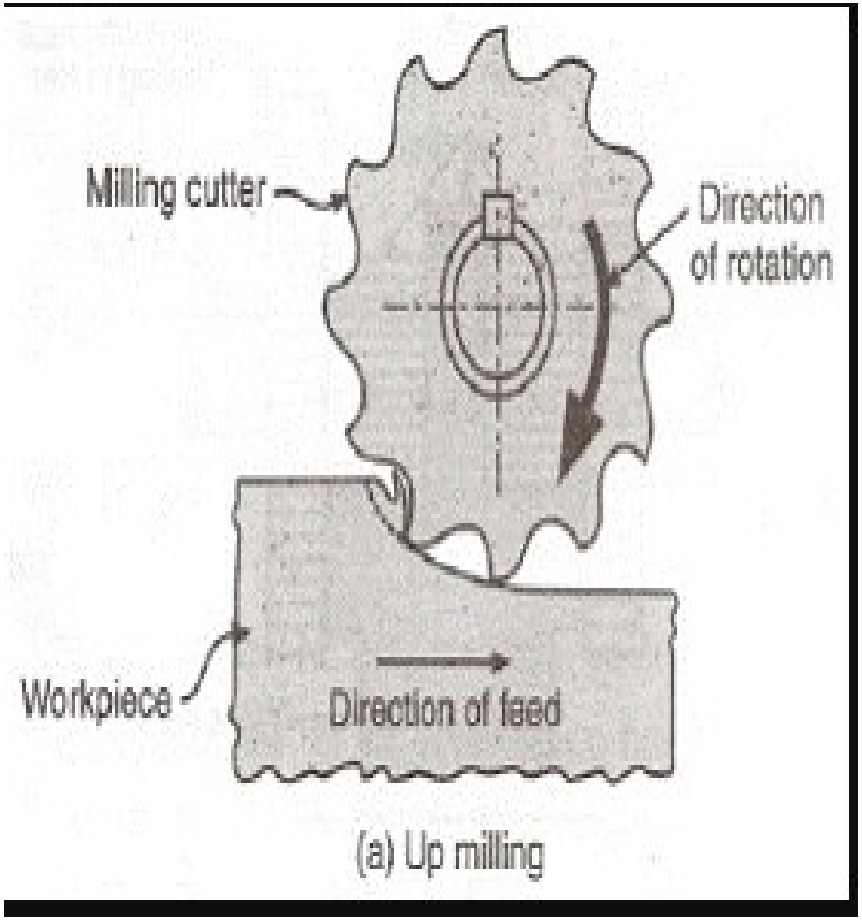
- Tapping is the operation for making internal threads in a hole by means of a tool called tap. The tap is essentially a bolt with threads cut on it.

Working principle of milling machine



- The workpiece is holding on the worktable of the machine. The table movement controls the feed of workpiece against the rotating cutter.
- The cutter is mounted on a spindle or arbor and revolves at high speed.
- As the workpiece advances, the cutter teeth remove the metal from the surface of workpiece and the desired shape is produced.

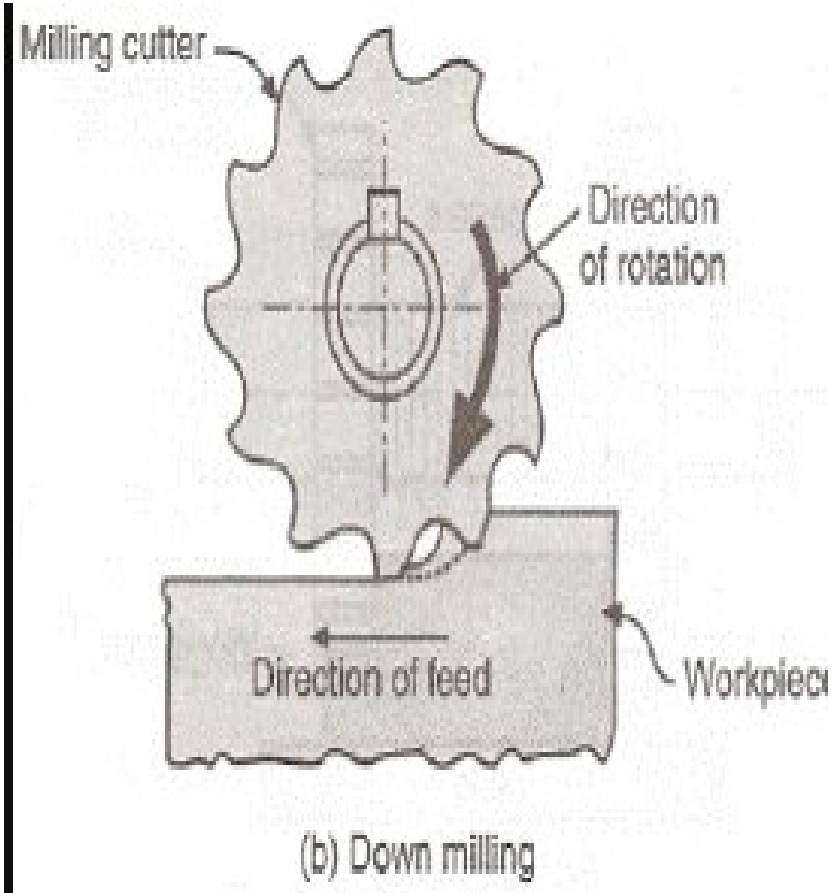
Up milling



Up milling:

- It is also called as **Conventional milling.**
- In which the workpiece is moving towards the cutter, opposing the cutter direction of rotation.
- The cutting force is minimum during the beginning of the cut and maximum at the end of the cut. In up milling, the cutting action tends to lift the workpiece and hence,
- A proper fixture is required in this operation.

Down milling

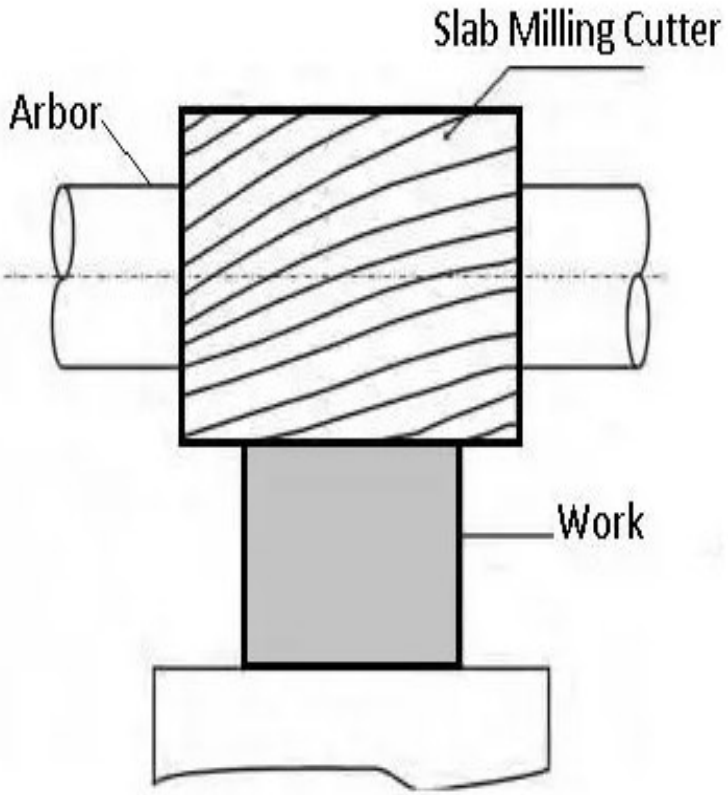


(b) Down milling

Down milling:

- It is also called as **Climb milling.**
- When the cutter rotation is in the same direction as the motion of the workpiece being fed.
- The cutting force is maximum at the beginning and minimum at the end of the cut.
- In down milling, the cutting force is directed on to the work table, which allows thinner parts to be machined without susceptibility to breakage.
- A better surface finish is obtained.

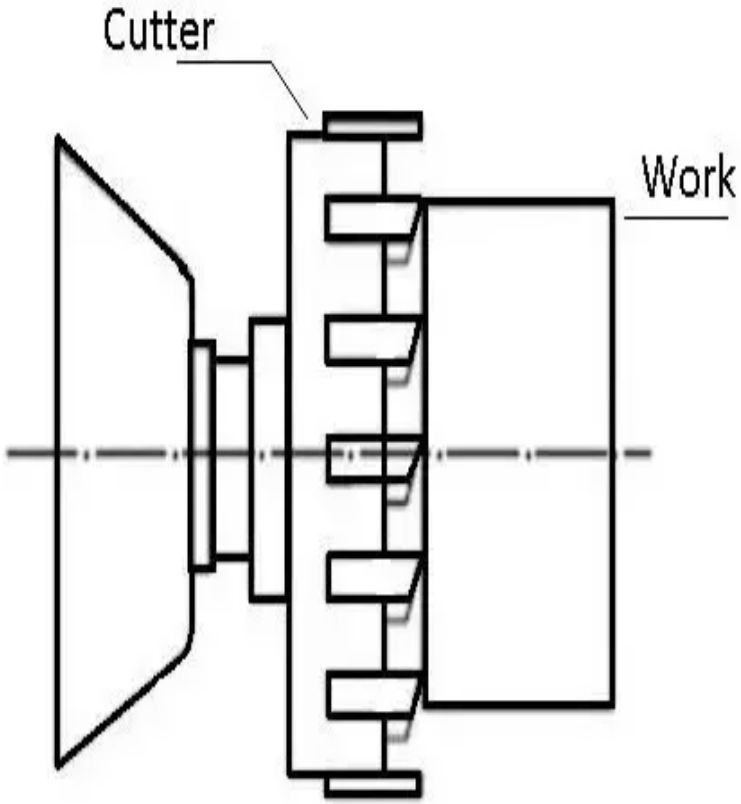
Plain milling



Plain Milling Operation

- Plain milling is the most common type of milling machine operation. Plain milling is performed to produce a plain, flat, horizontal surface parallel to the axis of rotation of a plain milling cutter.
- The operation is also known as slab milling. To operate, the work and the cutter are secured properly on the machine. The depth of cut is set by rotating the vertical feed screw of the table.

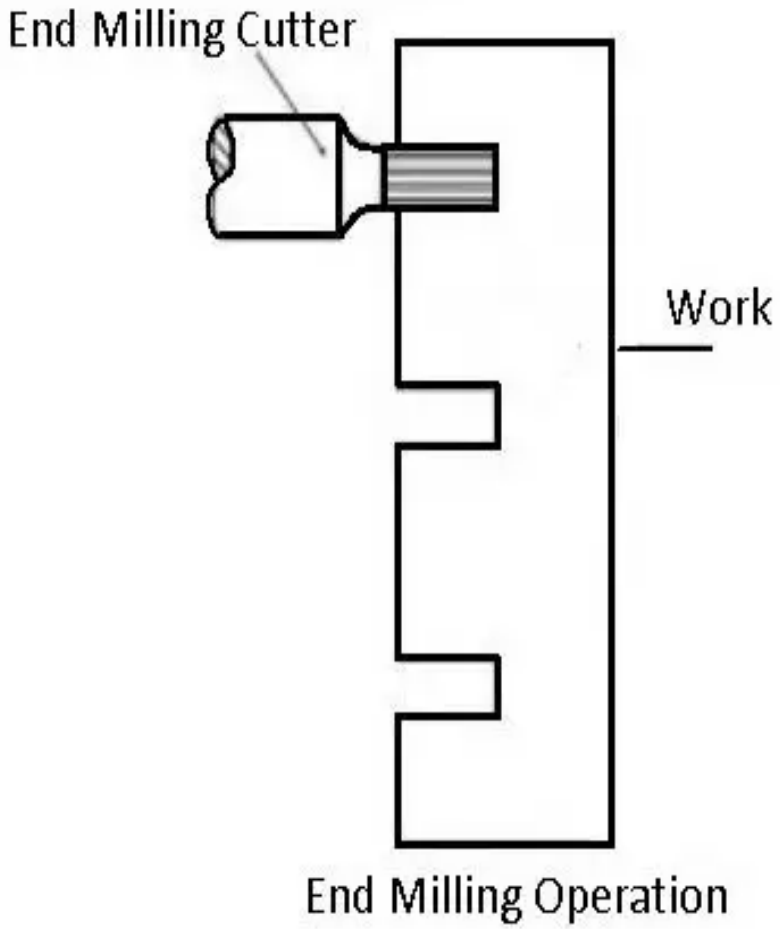
Face Milling



Face Milling Operation

- The face milling is the simplest milling machine operations. This operation is performed by a face milling cutter rotated about an axis perpendicular to the work surface.
- The operation is carried out in plain milling, and the cutter is mounted on a stub arbor to design a flat surface. The depth of cut is adjusted by rotating the cross feed screw of the table.

End milling



- End milling is the operation of producing a flat surface which may be vertical, horizontal, or at an angle about the table surface. The cutter used is an end mill.
- The end milling cutters are also used for the production of slots, grooves, or keyways.
- A vertical milling machine is more suitable for end milling operations.

Introduction of Advanced manufacturing

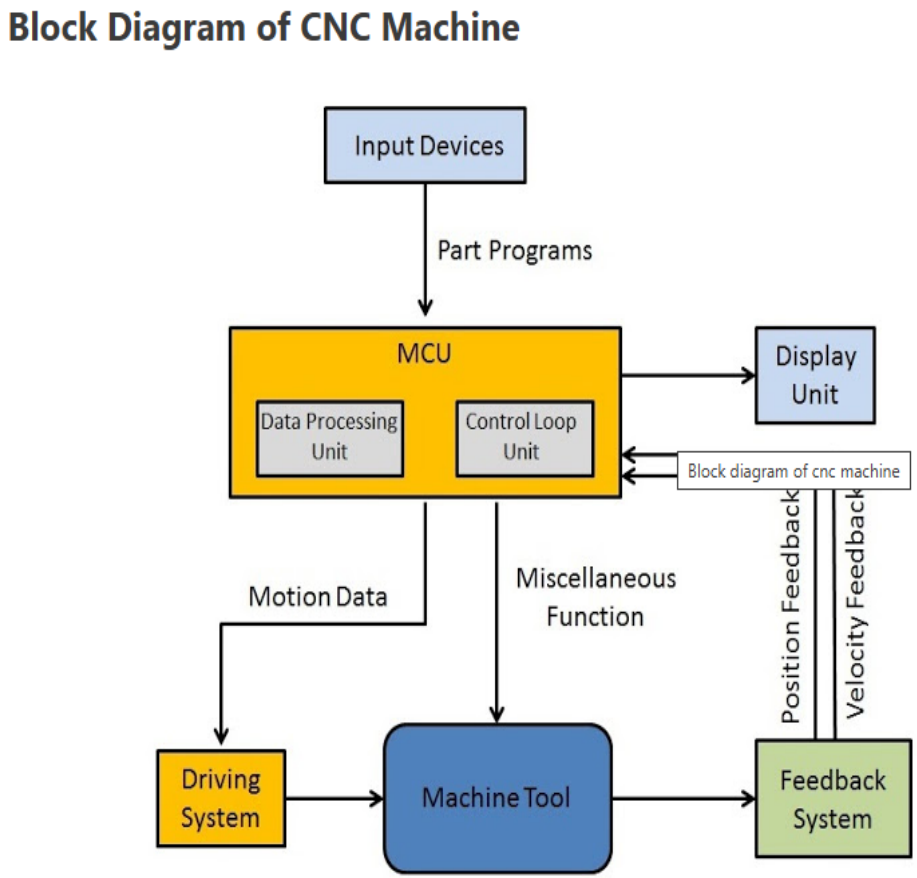
- Advanced manufacturing refers to the use of innovative technologies to create new products, refine existing products, and perform production activities that will improve the quality and process of manufacturing to give manufacturers a competitive edge.
- The purpose of advanced manufacturing is to increase output while optimizing value, quality, market responsiveness and flexibility.
- Advanced manufacturing also aims to reduce time to market, material inventory and content and unit quantities.
- Advanced Manufacturing Technology Examples
- Artificial intelligence
- Machine learning
- Robotics and automation
- Computer-Integrated Manufacturing (CIM)
- Flexible Manufacturing Systems (FMS)
- Additive Manufacturing (3D Printing)

Components of CNC

- CNC stands for Computer Numerical Control.
- When computers are used to control a Numerical Control (NC) machine tool than the machine is called CNC machine.
- In other words, the use of computers to control machine tools like lathe, drill, mills, shaper etc is called CNC machine.
- The cutting operations performed by the CNC is called CNC machining.

Components of CNC





Components of CNC

1) Input Devices: These are the devices which are used to input the part program in the CNC machine.
2) Machine Control Unit (MCU): It is the heart of the CNC machine. It consists of Data processing unit and control loop unit. It performs all the control.
3) Machine Tool: A CNC machine tool always has a slide table and a spindle to control of the position and speed.
4) Driving System: The driving system of a CNC machine consists of amplifier circuits, drive motors and ball lead screw.
5) Feedback System: This system consists of transducers that act as sensors. It is also called a measuring system.
6) Display Unit: A monitor is used to display the programs, commands and other useful data of CNC machine.

Advantages and applications of CNC

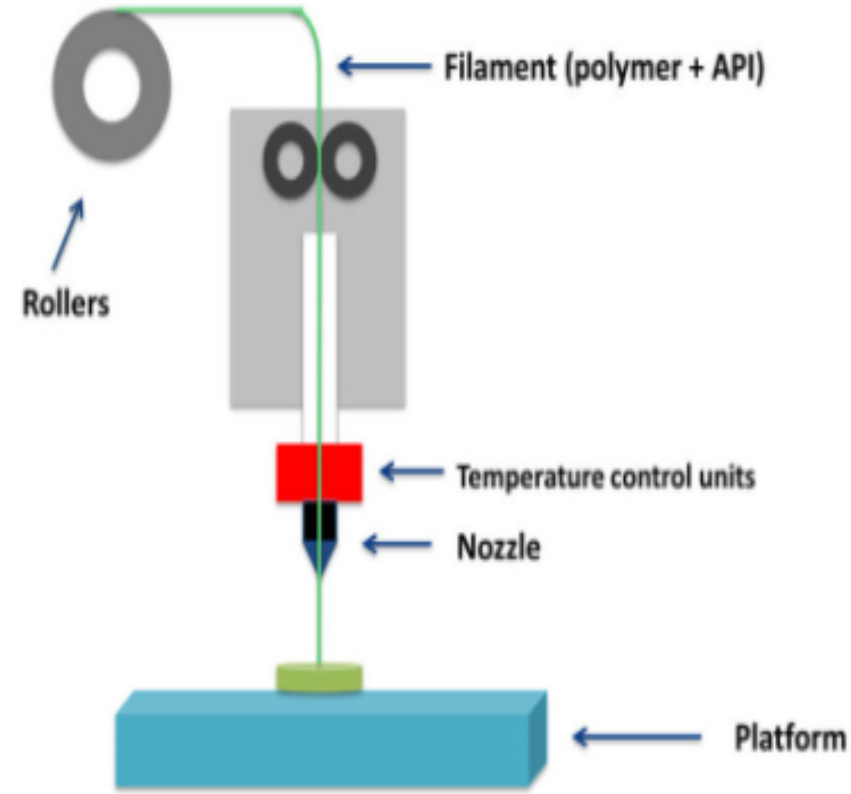
- Advantages of CNC Machines:

1) High Precision and Accuracy
2) Repeatability
3) Reduced Labor Costs
4) It can be run for 24 hours a day
5) Less Waste
6) Improved Safety

- Applications of CNC:

1) Aerospace Industry
2) Automotive Industry
3) Medical Equipment
4) Electronics
5) Woodworking
6) Metal Fabrication
7) Prototyping

3D printing



The diagram of FFF 3D printer

- 3D printing or additive manufacturing is a process of making three dimensional solid objects from a digital file.
- The 3D printing process builds a three-dimensional object from a computer-aided design (CAD) model, usually by successively adding material layer by layer, which is why it is also called additive manufacturing.
- 3D printing covers a variety of processes in, which material is joined or solidified under computer control to create a three-dimensional object, with material being added together typically layer by layer.